

ECEN721: Optical Interconnects Circuits and Systems Spring 2026

Lecture 10: Electroabsorption Modulator Transmitters



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Announcements

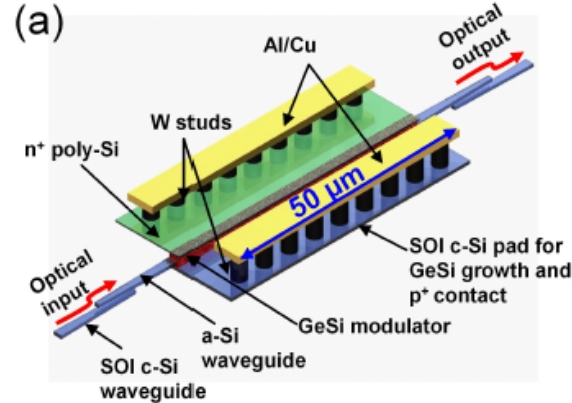
- Homework 3 due Mar 19
- Reading
 - Sackinger Chapter 8

Agenda

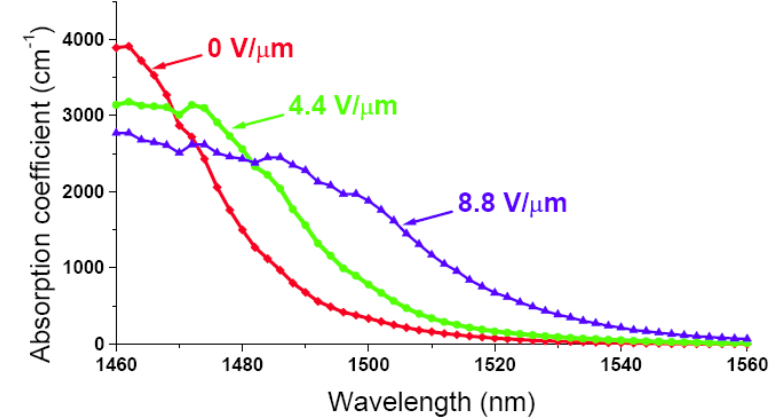
- EAM device operation and modeling
- EAM drivers
 - Controlled-impedance drivers
 - Lumped-element drivers

Electro-Absorption Modulator (EAM)

Waveguide EAM [Liu 2008]



[Helman JSTQE 2005]

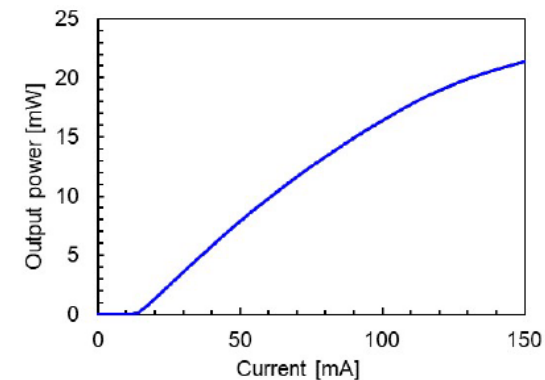
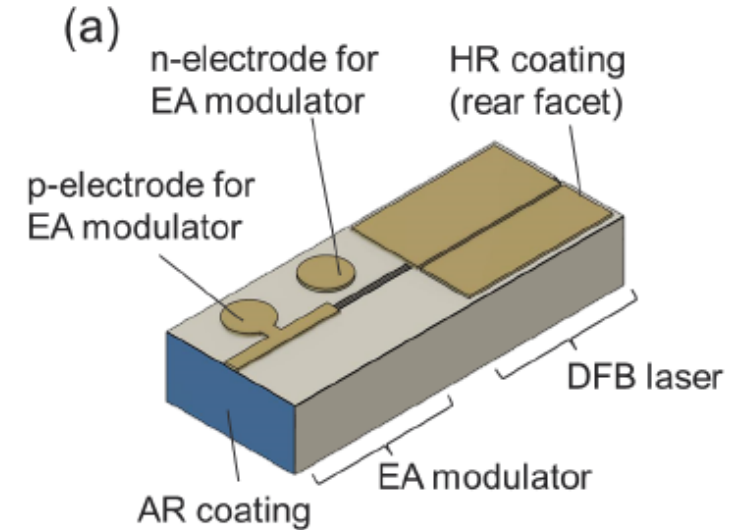


- Electro-absorption modulators operate with voltage-dependent absorption of light passing through the device
- The device structure is a reverse-biased p-i-n diode
- The Franz-Keldysh effect describes how the effective bandgap of the semiconductor decreases with increasing electric field, shifting the absorption edge
- While this effect is weak, it can be enhanced with device structures with multiple quantum wells (MQW) through the quantum-confined Stark effect

Monolithic DFB Laser + EAM (EA-DFB)

- InP platform allows for monolithic DFB laser and EAM integration
- DFB laser operates in forward bias and the EAM is in reverse bias
- 226Gb/s PAM4 operation achieved with 15-tap TX FFE
 - Semi-insulating substrate allows for 4 independent electrodes, 2 for DFB and 2 for EAM
 - Allows for differential drive of EAM
 - 5.1dB extinction ratio with $2V_{ppd}$ swing

[Ohno OFC 2025]

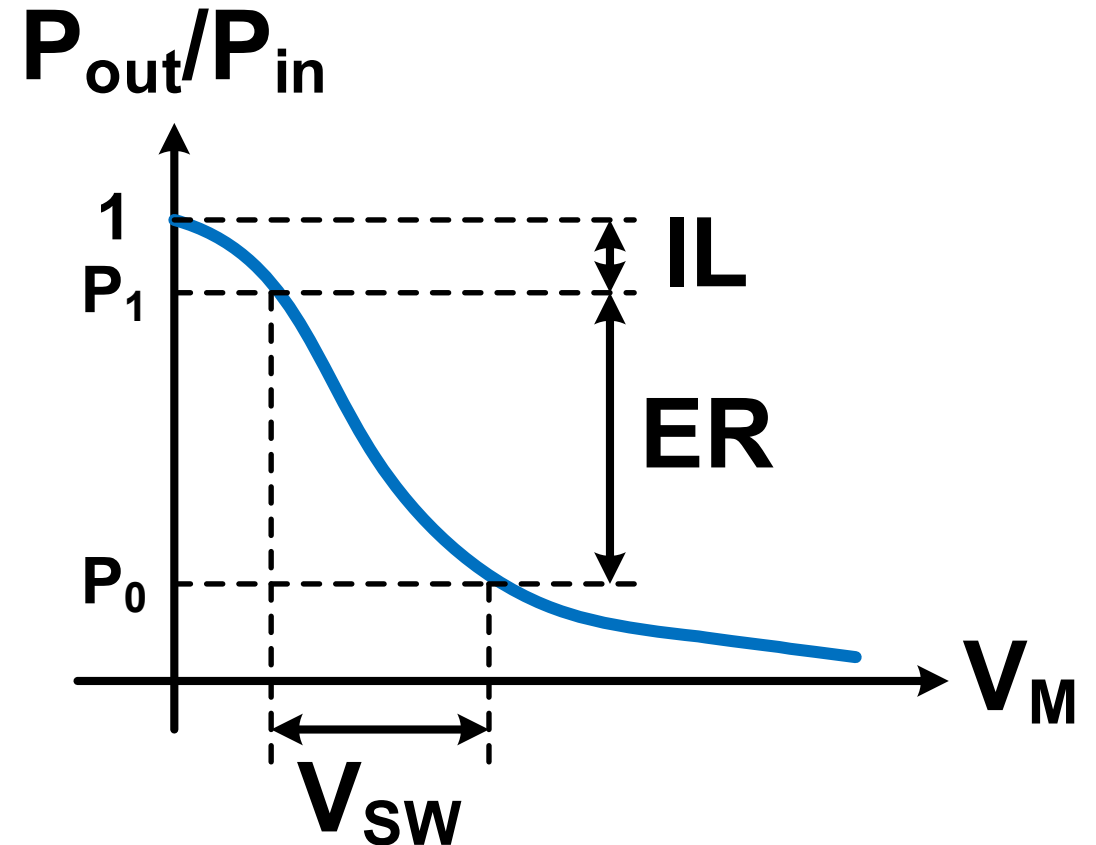


226Gb/s PAM4

Fiber length	BTB
OuterER [dB]	4.8
TDECQ [dB]	2.13
Waveform with 15-tap FFE	

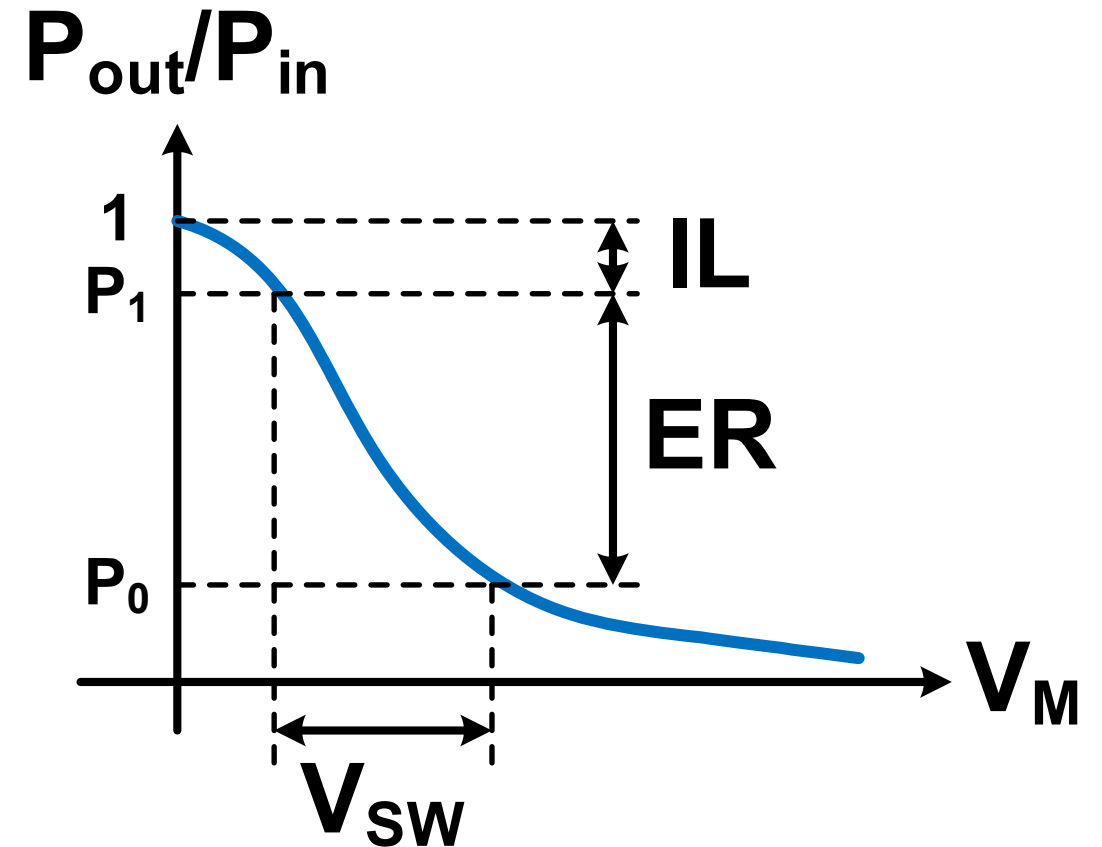
EAM Switching Curve

- At low reverse-bias, the device ideally has low absorption and most of the light appears at the output
- The absorption increases when a strong reverse-bias is applied and less power appears at the output
- EAMs are characterized with a switching voltage V_{sw} that corresponds to a given extinction ratio
- Typical switching voltages are 1.5 to 4V



EAM Chirp

- The modulation voltage not only changes the absorption, but also the refractive index, inducing some chirp in the EAM output
- This chirp is generally much less than a directly-modulated laser, with $|\alpha| < 1$
- Application of a small on-state bias (0-1V) can minimize this chirp at the cost of higher insertion loss



EAM Equivalent Model

[Deshours JLT 2011]

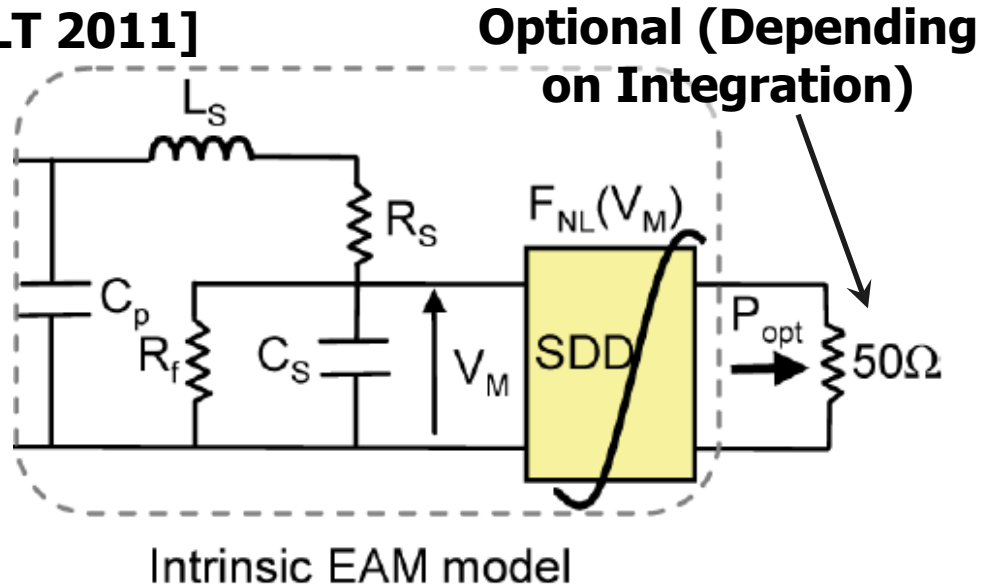


TABLE I
EXTRACTED PARAMETERS COMPONENTS OF THE EAM EQUIVALENT CIRCUIT

Symbol	Parameter	Value
R_s	serial resistance	12 Ω
C_s	junction capacitance	90 fF
R_f	leakage resistance	50 k Ω
C_p	bond-pad capacitance	20 fF
L_s	parasitic metallization inductance	10 pH

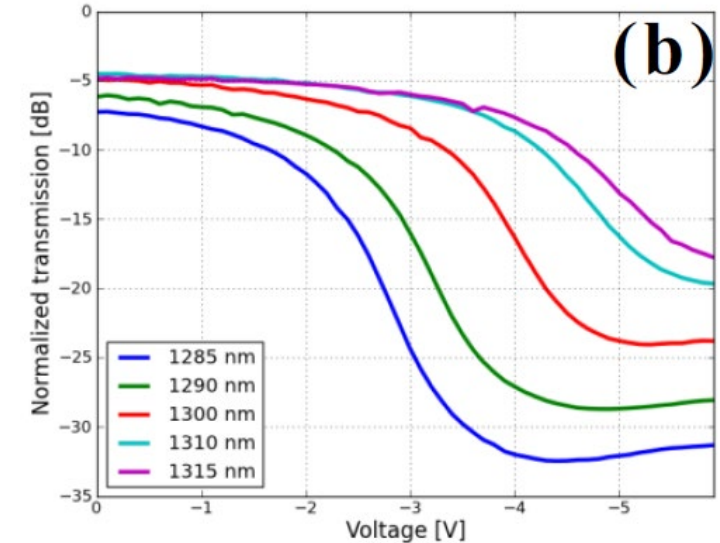
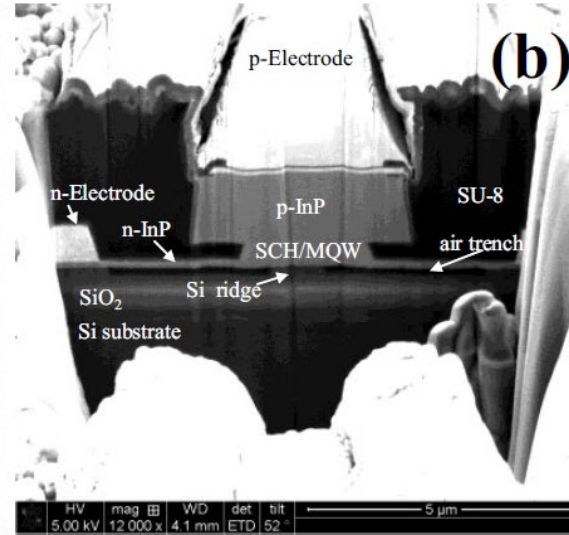
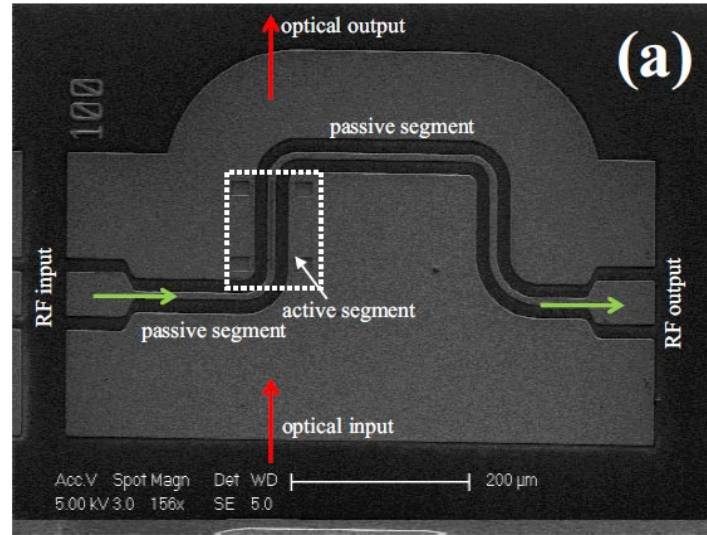
$$F_{NL}(V_M) = \frac{A_0 + \sum_{k=1}^6 A_k \times V_M^k}{1 + \sum_{k=1}^4 B_k \times V_M^k}$$

Extracted from curve fitting measurements

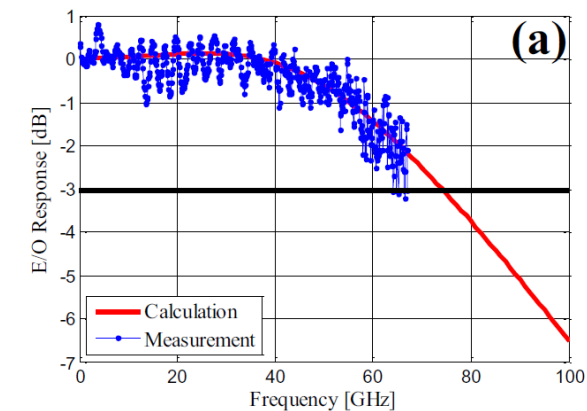
- Electrically, the EAM is a reverse-based diode
- This is modeled with a reverse-bias diode model and a non-linear absorption block $F_{NL}(V_M)$
- Depending on the integration level with the driver, the device may also include a termination resistor

67GHz Hybrid Silicon (InP) EAM

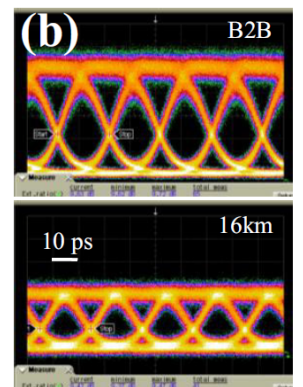
[Tang OFC 2012]



- EAM is formed with an InP p-i-n diode bonded onto silicon
- Design for a controlled-impedance driver
- Nominal 1300nm operation with -4V bias and 2V drive achieves $\sim 15\text{dB}$ ER

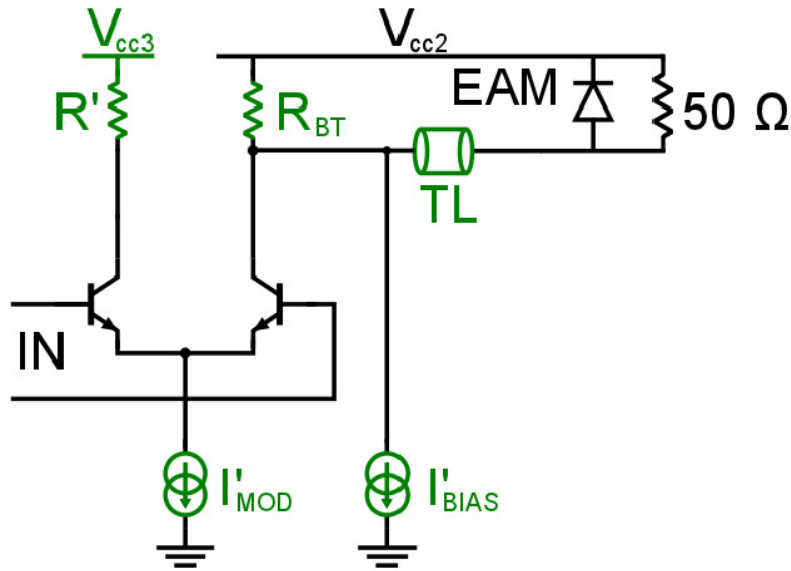


50Gb/s

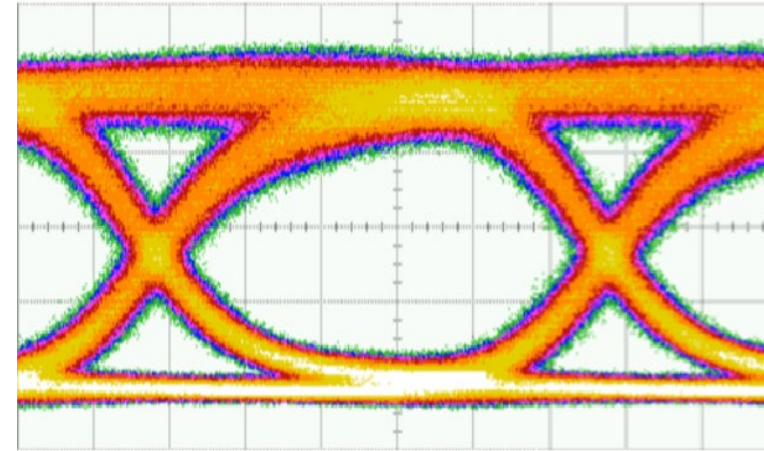


Controlled-Impedance EAM Driver

[Vaernewyck Opt. Exp. 2013]

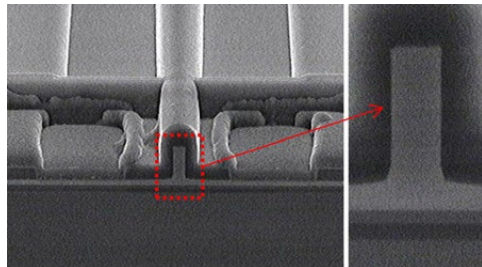
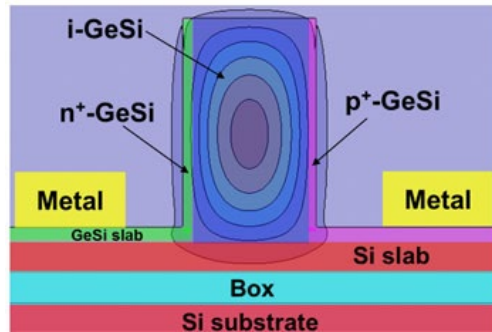
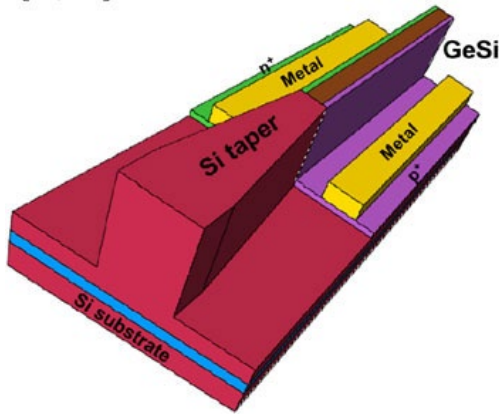


10Gb/s w/ 2.5V swing
& -1.7V bias

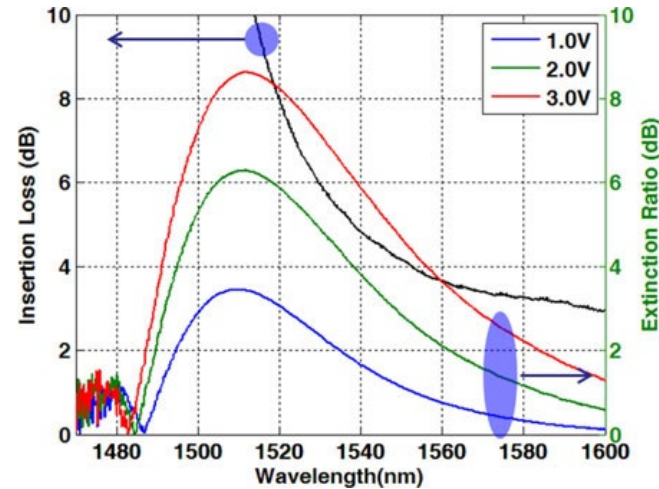


- If the EAM is not tightly integrated with the driver circuitry, then a controlled-impedance driver is required
- Bias current used to set operating point
- The high EAM swings results in large power consumption

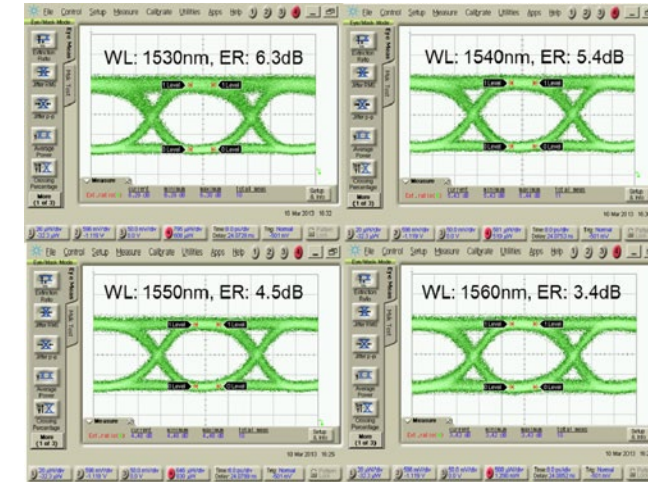
28Gb/s GeSi EAM on SOI



[Feng JSTQE 2013]



28Gb/s w/ 3V Swing

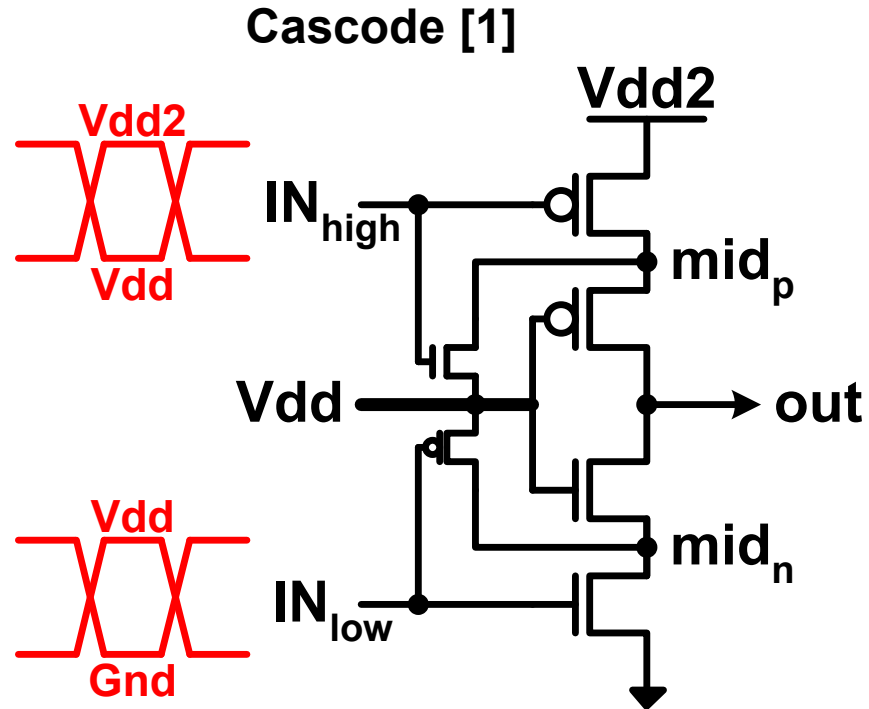


- EAM is formed with an GeSi p-i-n diode fabricated in an SOI platform
- Device is only 50um long and can be driven with a lumped-element driver
- Nominal operation with 3V drive achieves 3-6dB ER over a wide wavelength range

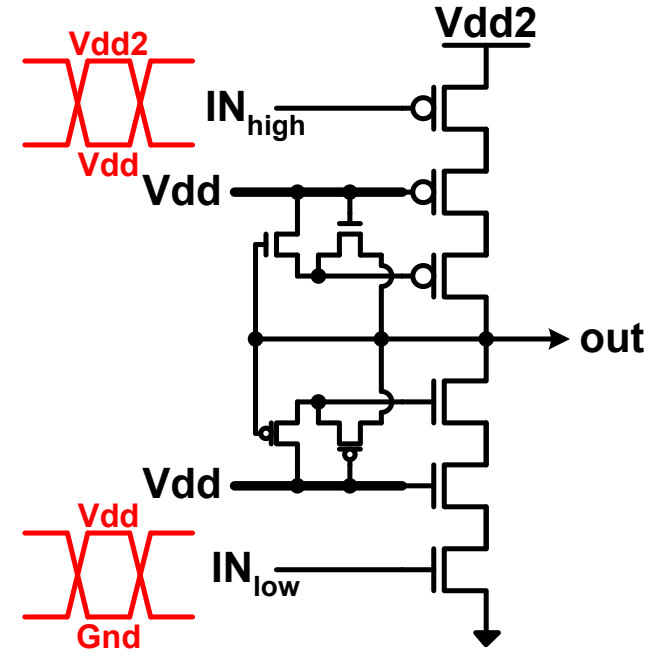
CMOS Reliability Constraints

- High electric fields in modern CMOS devices cause many reliability issues
 - Oxide Breakdown
 - Hot-Carrier Degradation
- Higher voltage I/O transistors are too slow
- Core transistor output stage V_{GS} , V_{GD} , V_{DS}
 - Should not exceed 20-30% of nominal V_{DD} during transients
 - Not greater than V_{DD} in steady state

High-Voltage Output Stages



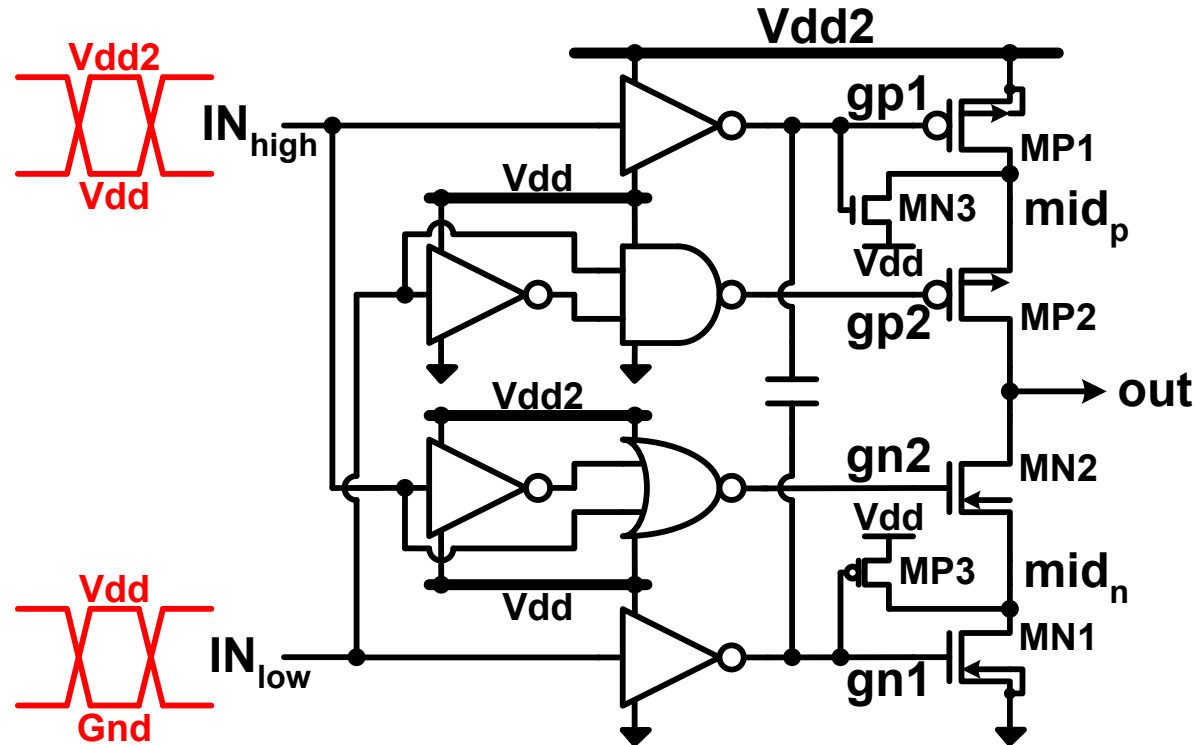
Double Cascode w/ Output Tracking [2]



- Static-biased cascode suffers from V_{DS} stress during transients
- Double-cascode with output tracking is slow due to three transistor stack and feedback loop

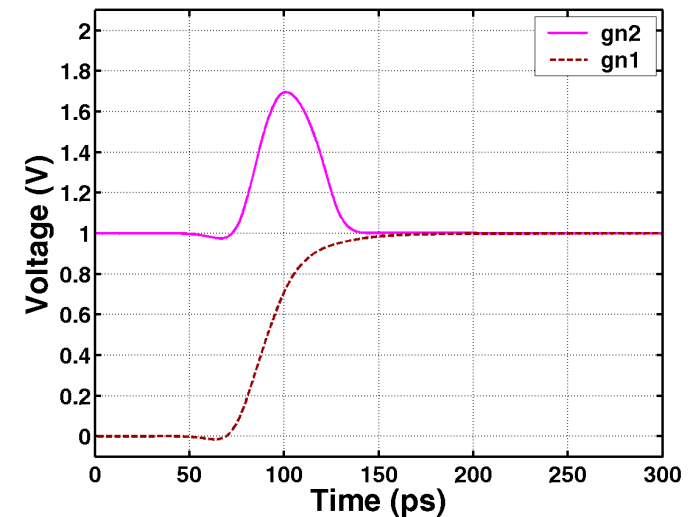
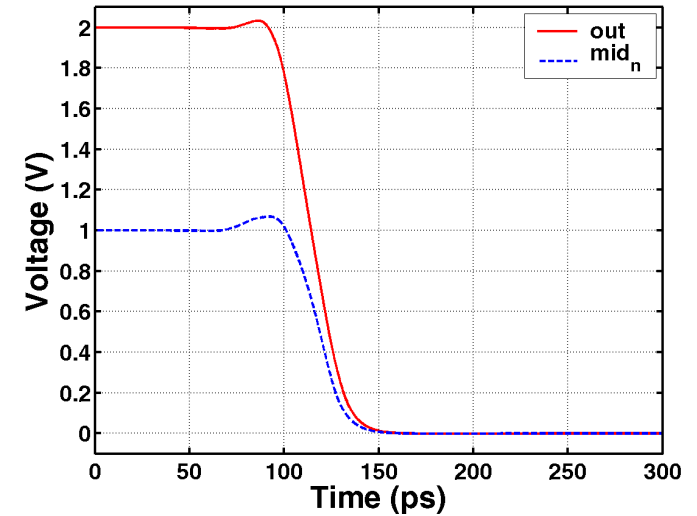
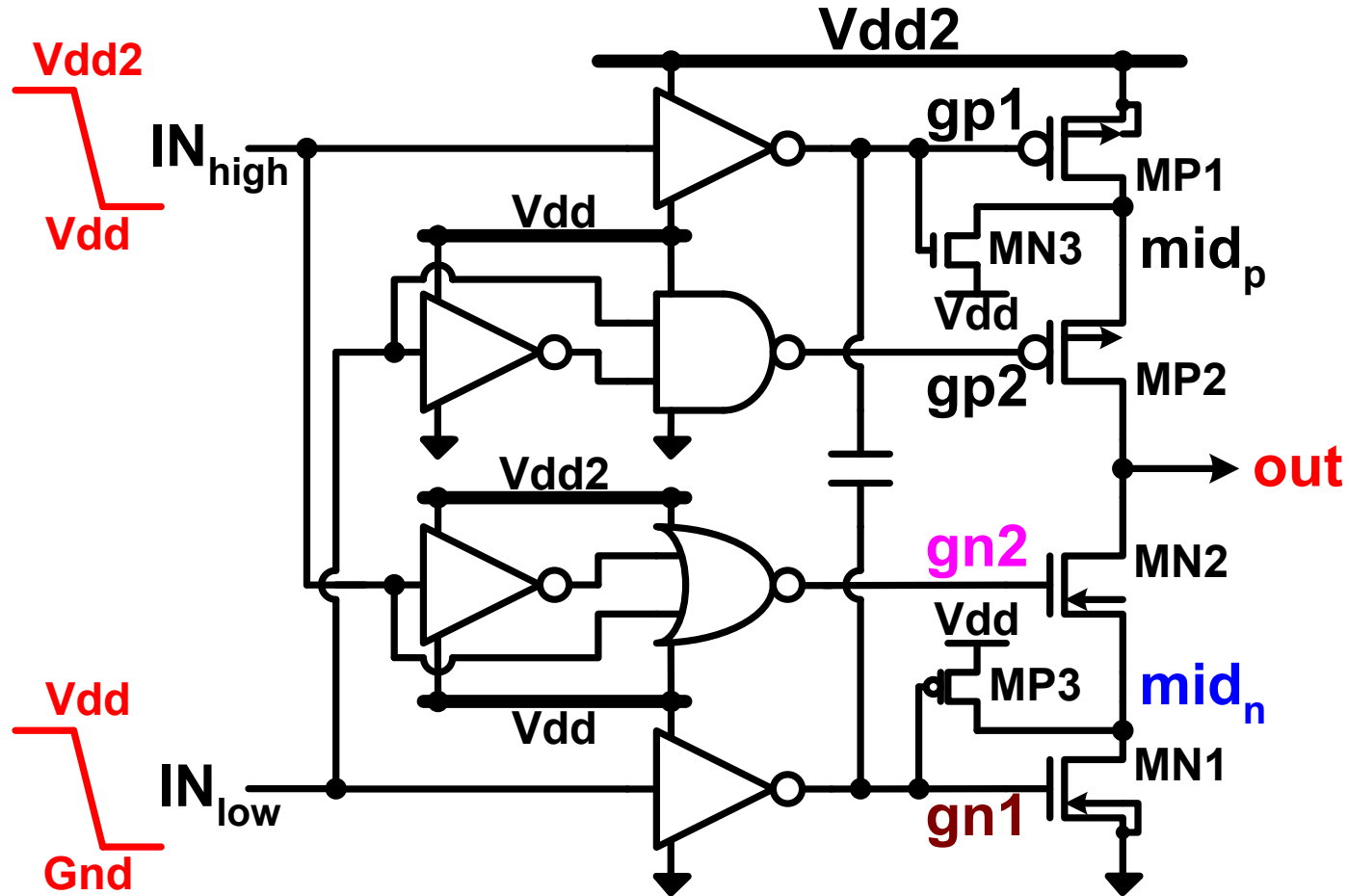
1. T. Woodward *et al*, "Modulator-Driver Circuits for Optoelectronic VLSI," *IEEE Photonics Technology Letters*, June 1997.
2. A. Annema *et al*, "5.5-V I/O in a 2.5-V 0.25- μ m CMOS Technology," *IEEE Journal of Solid-State Circuits*, Mar. 2001.

Pulsed-Cascode Output Stage

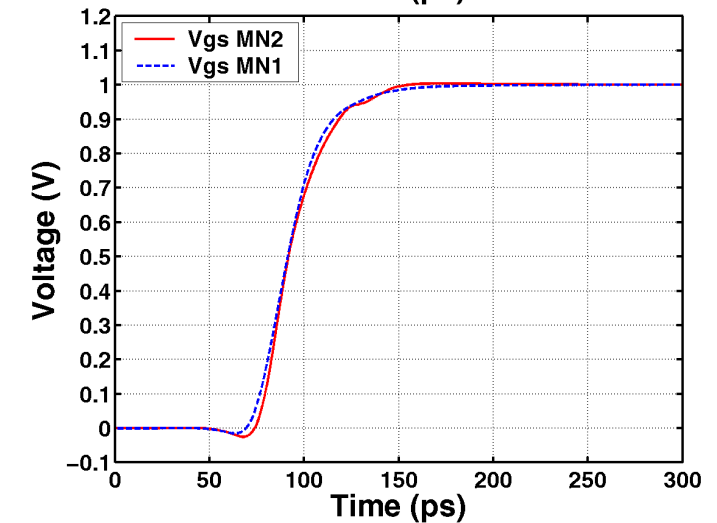
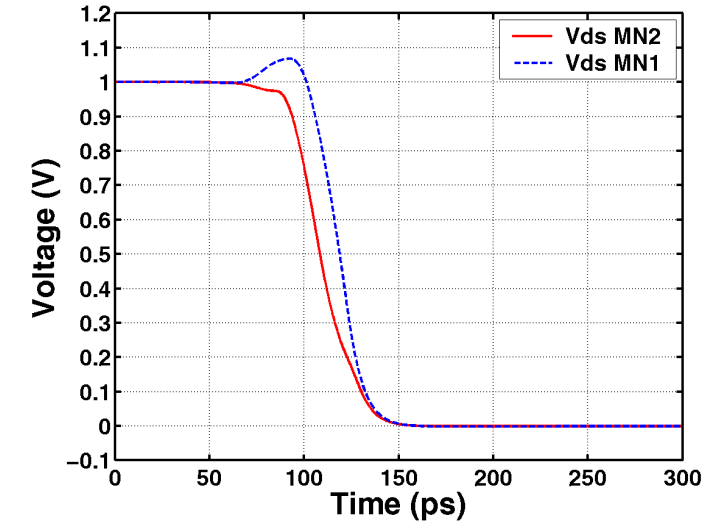
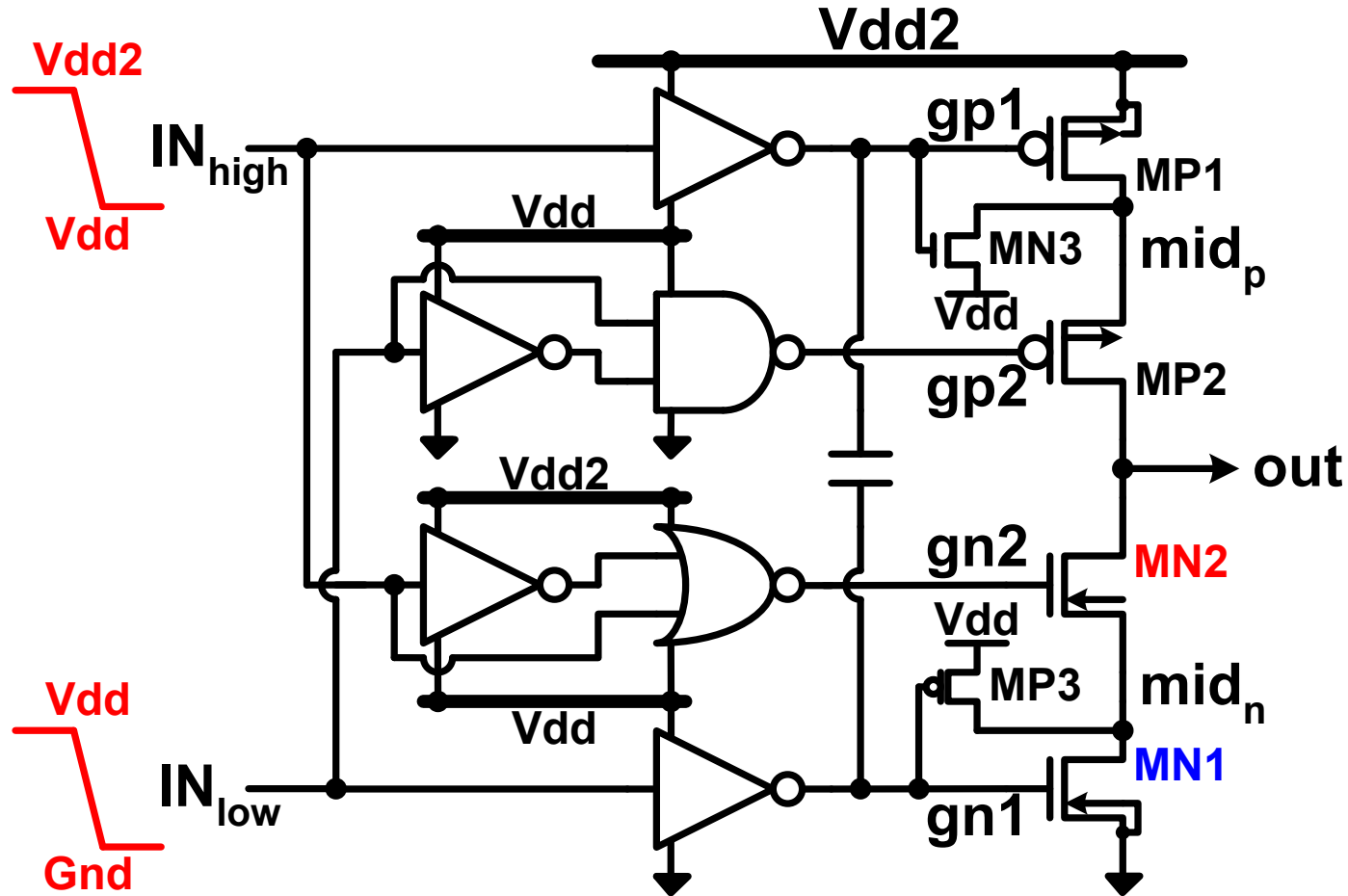


- Uses only two-transistor stack for maximum speed
- The cascode transistors gates are pulsed during a transition to prevent V_{ds} overstress

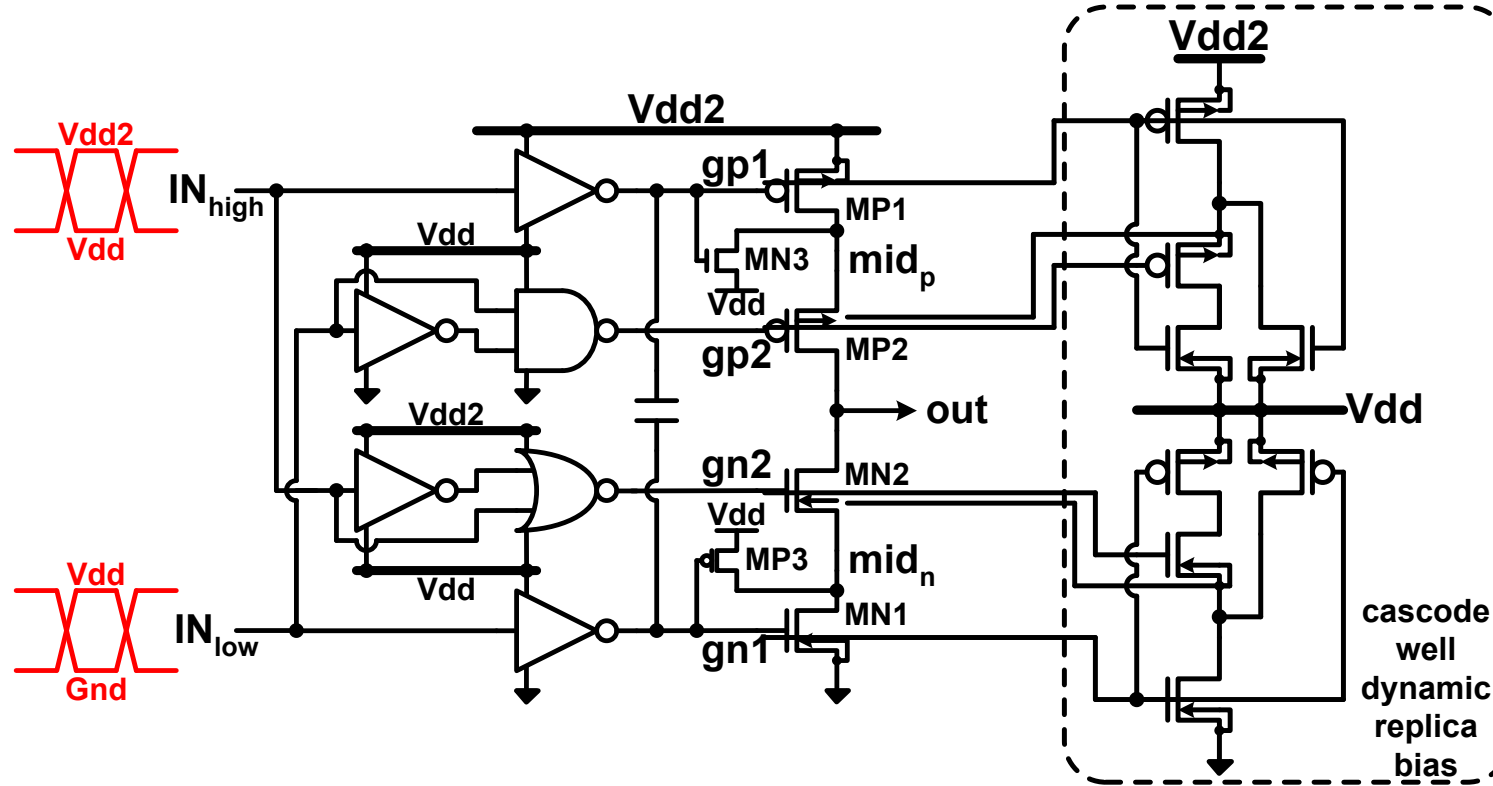
Output Stage Waveforms



Output Stage Waveforms

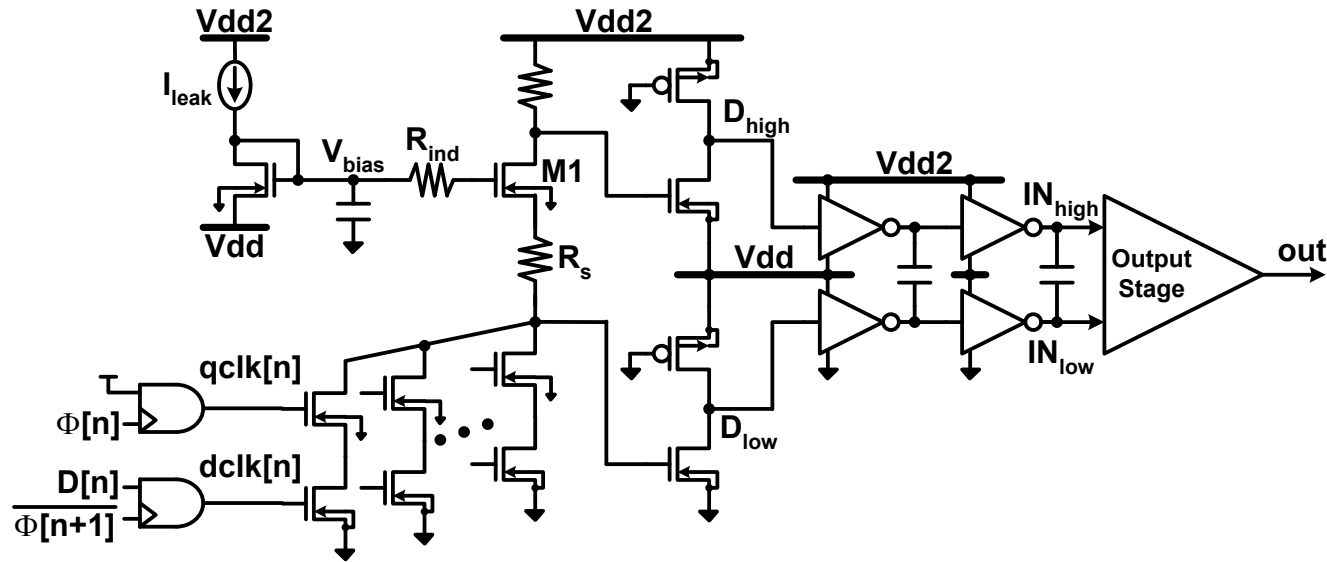


Pulsed-Cascode Output Stage



- Cascode body terminals dynamically biased to minimize body effect
- Issues
 - Voltage level shift
 - Delay matching high voltage path with standard voltage path

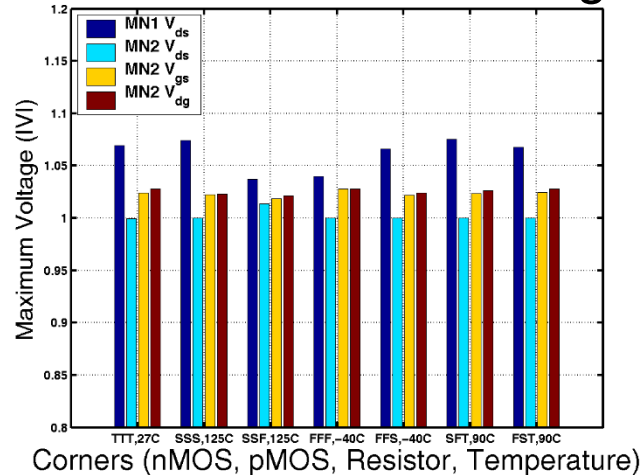
Modulator TX with Level-Shifting Multiplexer



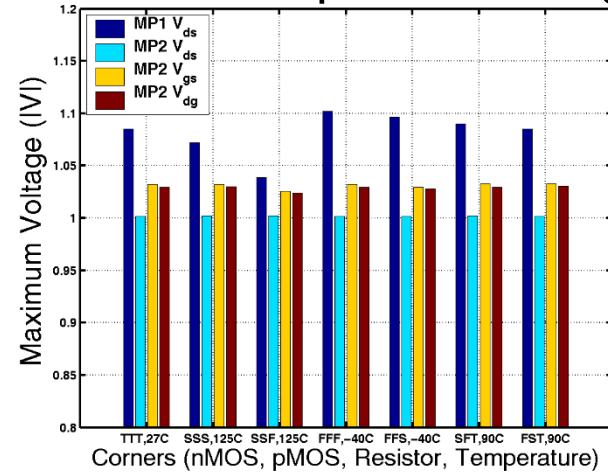
- Level-shifter combined with multiplexer
- Active inductive shunt peaking compensates multiplexer self-loading (reduces risetime by 37%)
- Slightly lower fan-out ratio in “high” signal path to compensate for level-shifting delay
- Delay Tracking
 - “High” path inverter nMOS in separate p-well
 - Metal fringe coupling capacitors perform skew compensation

Modulator Driver Reliability Simulations

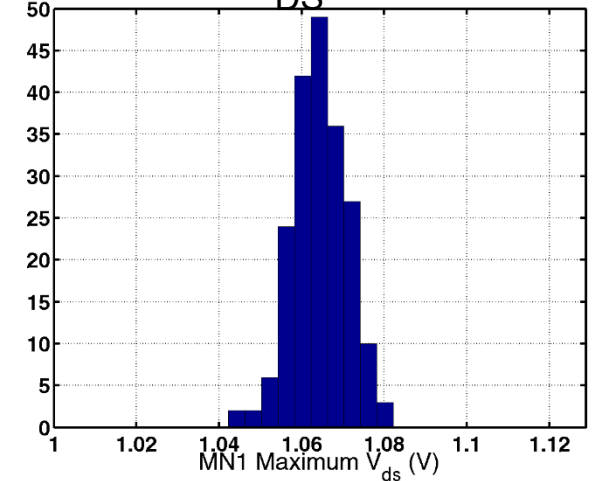
Maximum nMOS Voltages



Maximum pMOS Voltages

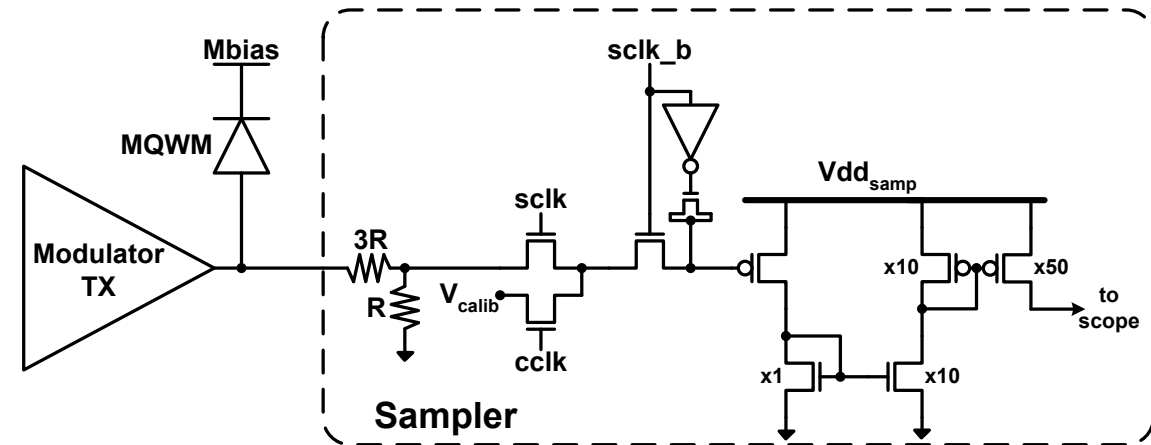
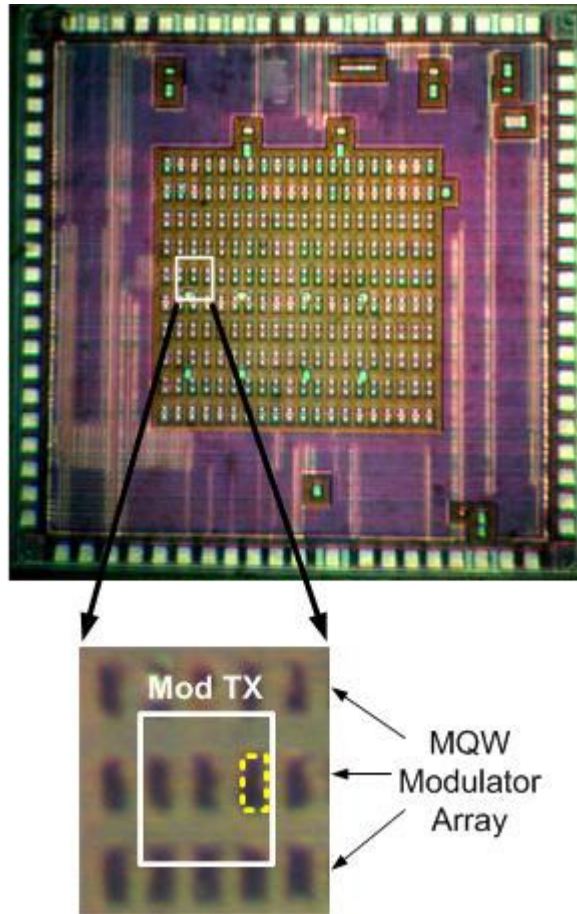


MN1 V_{DS} Distribution



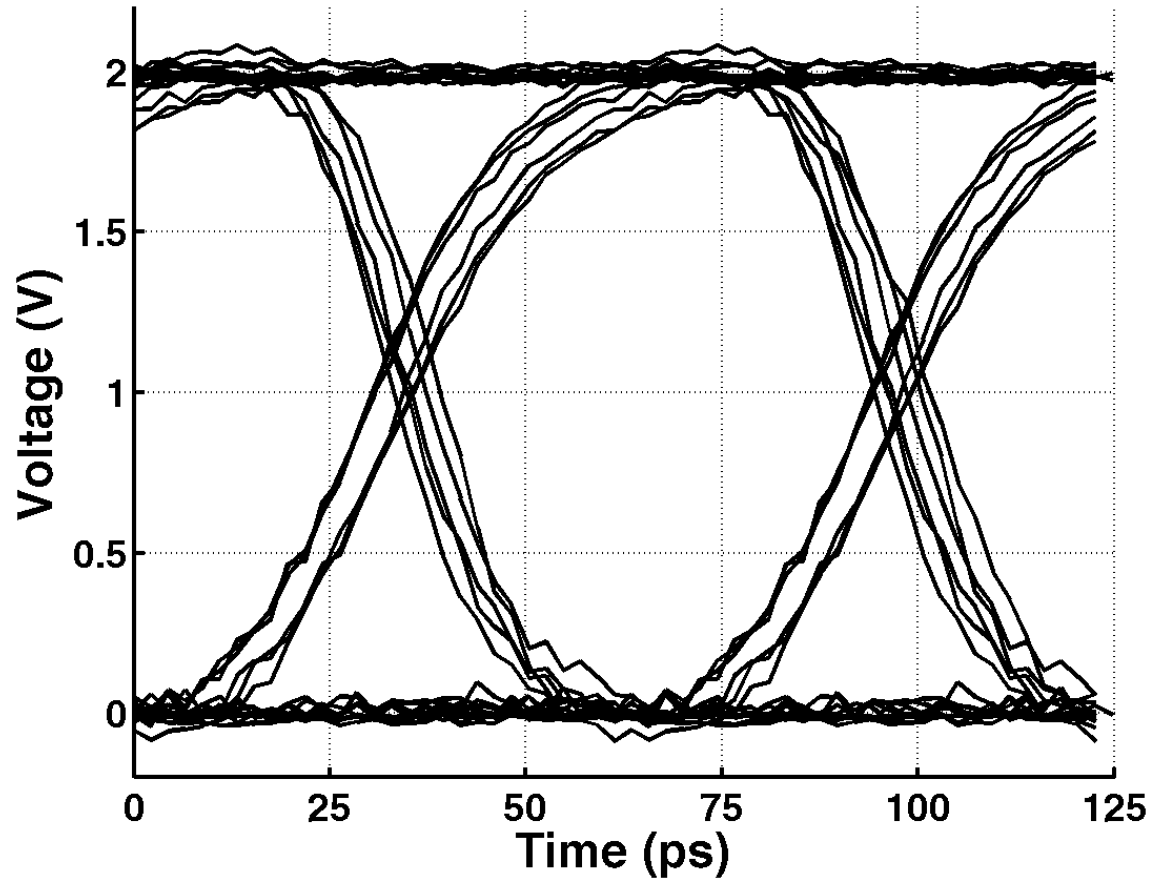
- Transient with random data
- Corner simulations show no output stage voltages exceed 11% of nominal Vdd
- Monte Carlo simulations show tight distributions ($\sigma < 15\text{mV}$)

MQWM TX Testing



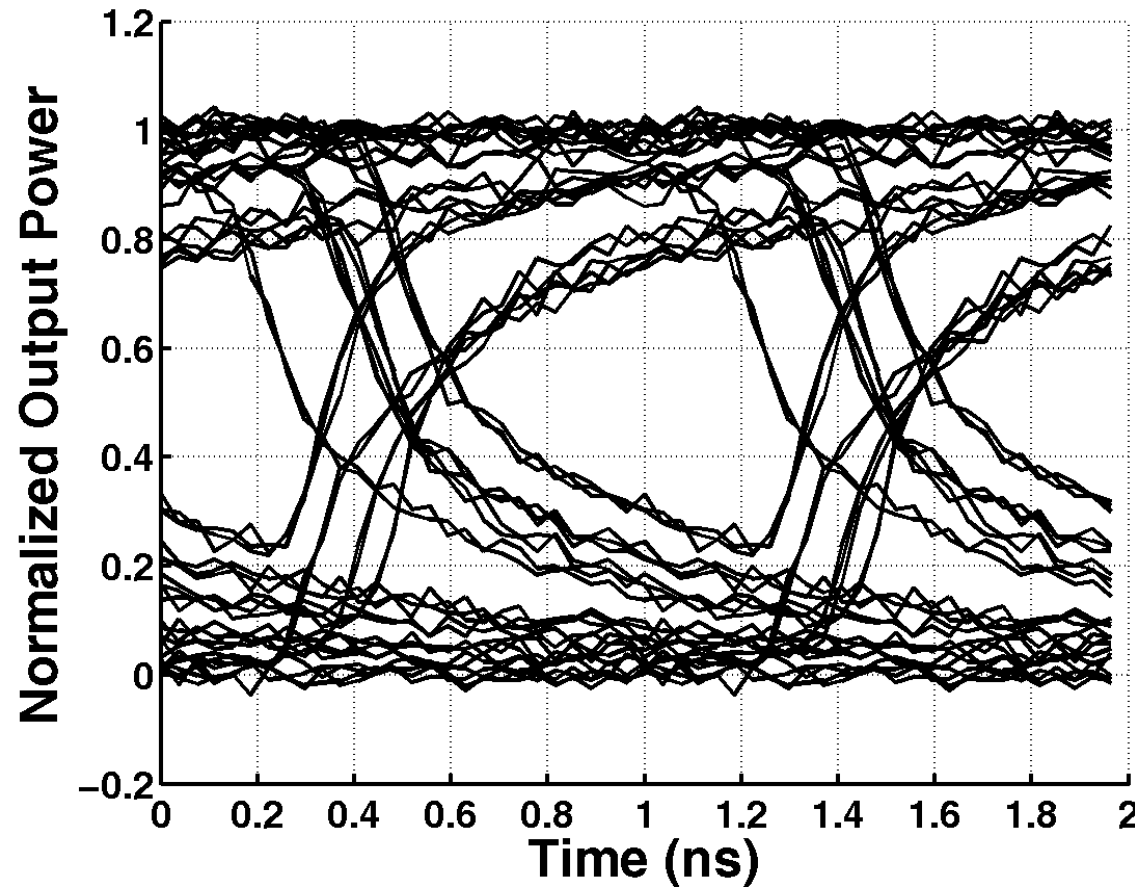
Electrical sampler at modulator transmitter output

Modulator Driver Electrical Eye Diagram



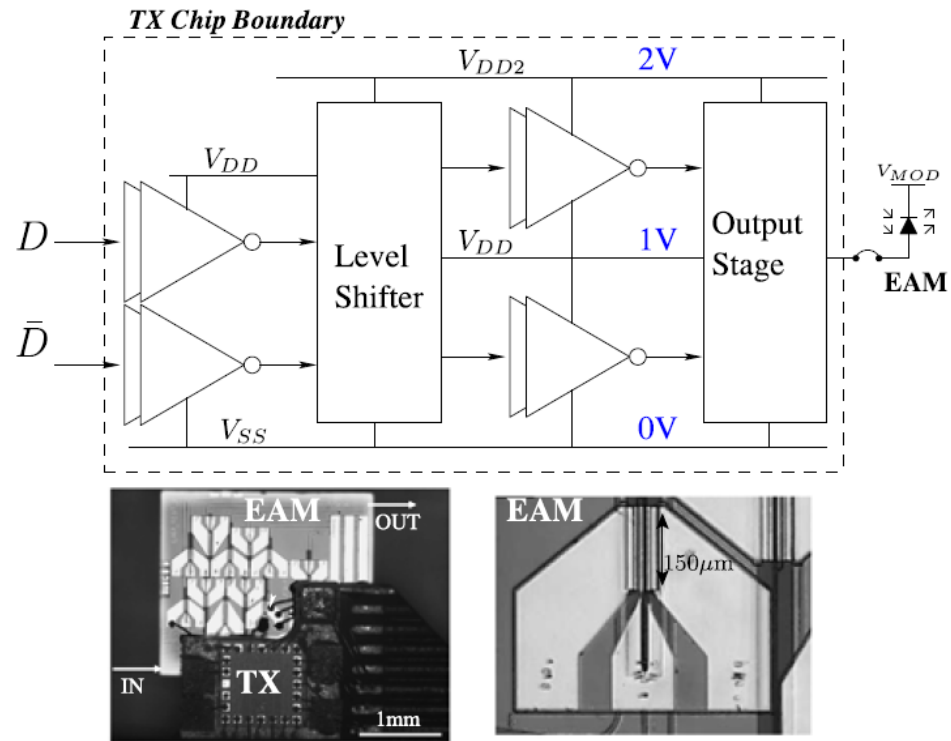
- 16Gb/s data subsampled at modulator driver output node

Modulator Driver Optical Eye Diagram

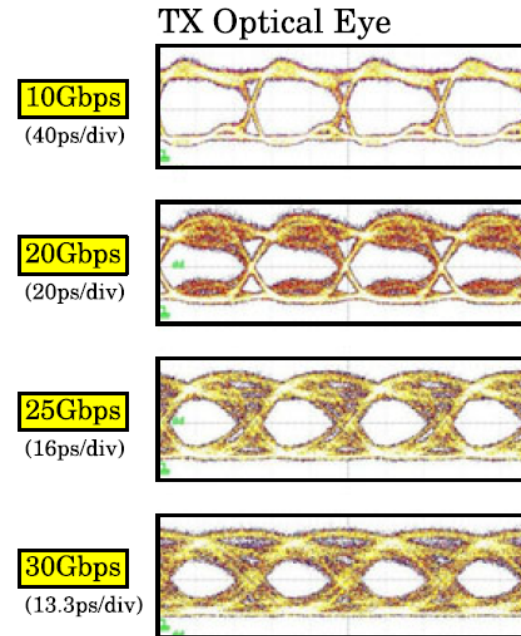


- Optical performance limited to $\sim 1\text{Gb/s}$ by poor modulator contact design causing large series resistance

30Gb/s Lumped-Element EAM Driver



[Dupuis JLT 2015]



- Using a 5.4V reverse bias and 2Vpp dynamic swing to achieve 8dB ER
- Have ~ 7 dB insertion loss

Next Time

- Ring Resonator Modulator (RRM) TX